

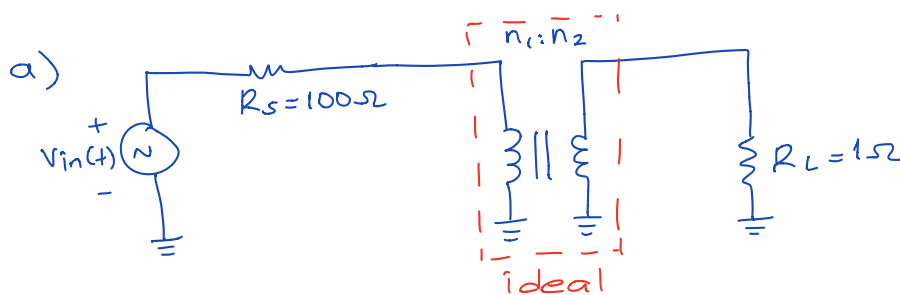
Name:

Student ID:

Duration: 20 minutes

We have a source at $\omega_0 = 10^7$ rps with a source voltage of $v_{in}(t) = 10 \cos(\omega_0 t)$ V and a source impedance of $R_s = 100 \Omega$. We want to deliver maximum power to a load resistance of $R_L = 1 \Omega$.

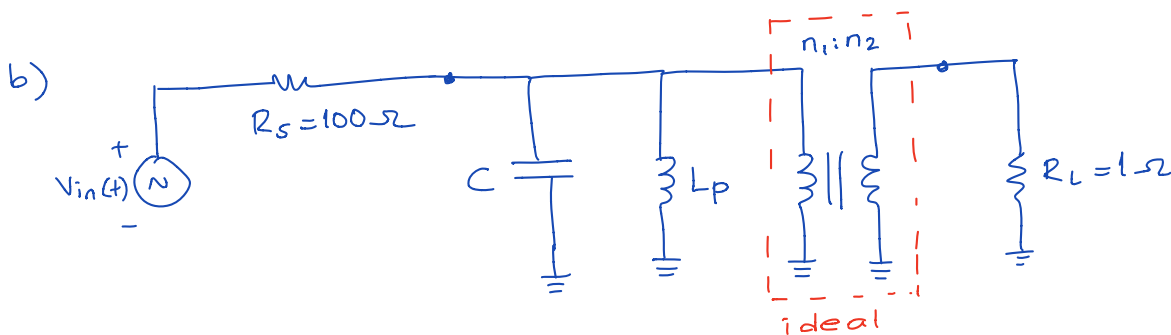
- a) (30 pts) Use an ideal transformer placed between the source and the load to deliver maximum power at ω_0 . What should be turns ratio ($n_1:n_2$) of the transformer? Draw the overall circuit.
- b) (60 pts) Now, use the more realistic transformer model that considers inductance of the primary, L_p . Choose the number of turns for the primary and the secondary. What is L_p if we use a core with an inductance constant $A_L = 100$ nH/turns²? Find the value of the additional component needed for impedance matching. Draw the overall circuit. **Hint:** For inductors, $L = n^2 A_L$.
- c) (10 pts) For this source, what is the maximum power that can be delivered to a load?



$$R_s = \left(\frac{n_1}{n_2}\right)^2 \cdot R_L$$

$$100 = \left(\frac{n_1}{n_2}\right)^2 \cdot 1$$

$$\boxed{\frac{n_1}{n_2} = 10}$$



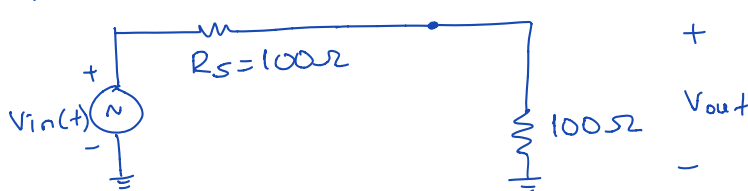
Choosing $\boxed{n_1 = 10, n_2 = 1}$

$$L_p = n_1^2 A_L = 100 \times 100 \text{ nH} = \boxed{10 \mu\text{H}}$$

$$\omega_0 = \frac{1}{\sqrt{C L_p}} \Rightarrow C = \frac{1}{\omega_0^2 L_p} = \frac{1}{10^{14} \cdot 10 \cdot 10^{-6}} = 10^{-9} \text{ F} = \boxed{1 \text{ nF}}$$

Note: Other n_1 & n_2 can be chosen, as well. For example:
 $n_1 = 20, n_2 = 2 \Rightarrow L_p = 40 \mu\text{H}, C = 0.25 \text{ nF}$

c) Circuit seen by the source:



$$V_{out}(t) = \frac{V_{in}}{2} = 5 \cos(\omega_0 t) \text{ V}$$

$$P_{L, \max} = \frac{5^2}{2} \cdot \frac{1}{100} = \boxed{\frac{1}{8} \text{ W}}$$

Alternatively: $P_{L, \max} = \frac{|V_s|^2}{8 R_s} = \frac{100}{8 \cdot 100} = \frac{1}{8} \text{ W}$